FIELD INSTRUMENTATION / RADIO SCIENCE

Project SpaceRF:
An Ionospheric Solar Flare & Cosmic Radiation Monitor

Repurposing mobile audio subsystems into a portable VLF software-defined-radio (SDR) receiver frontend.

REFERENCE CARRIER

18.20 kHz

VTX3 • South Vijayanarayanam, Tamil Nadu, India

VISUAL CONCEPT / LIVE MONITORING UI



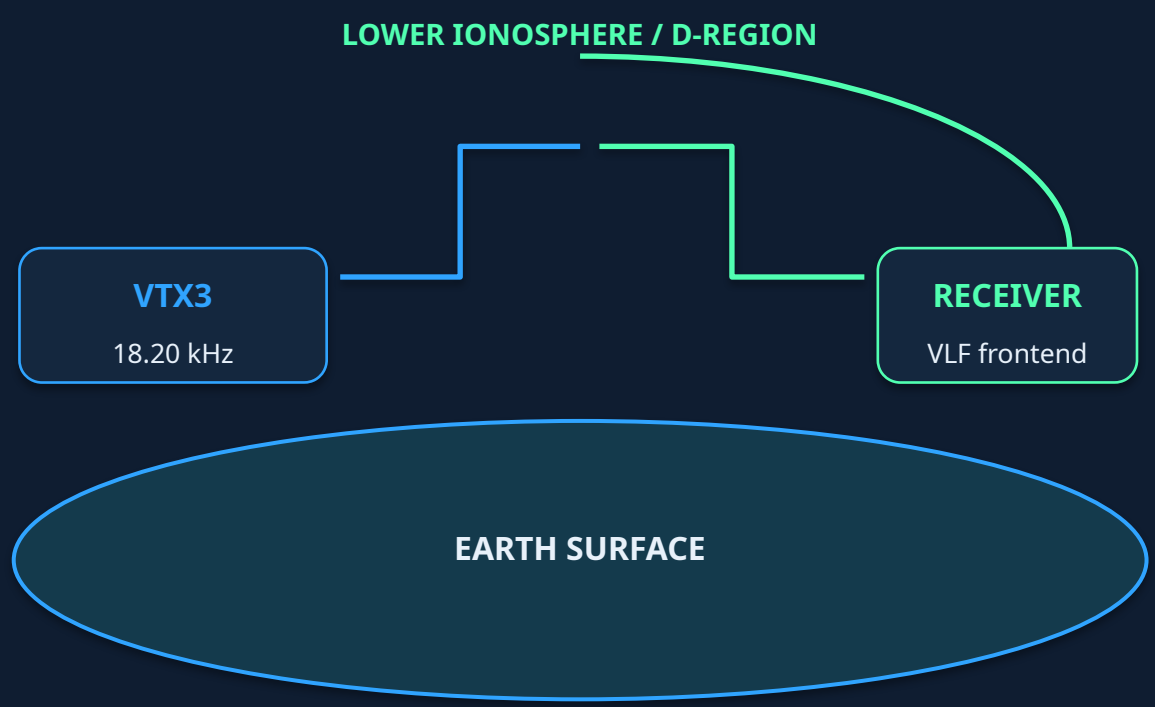
Illustrative dashboard — not measured data

DESIGN THESIS • LOW-COST RF FRONTEND + EXISTING PHONE DSP

Scientific Principles | Earth–Ionosphere Waveguide

PROPAGATION MODEL

PLANET-SCALE PROPAGATION PATH



Simplified propagation illustration; not to scale.

- **Solar forcing:** UV and X-ray radiation ionize atmospheric constituents.
- **Waveguide:** Conductive lower ionosphere and ground guide VLF energy over long paths.
- **Carrier:** VTX3 at South Vijayanarayanam is listed at 18.2 kHz.
- **Observable:** Changes in received amplitude/phase can reveal path perturbations.

VLF = 3–30 kHz • λ at 18.2 kHz ≈ 16.5 km

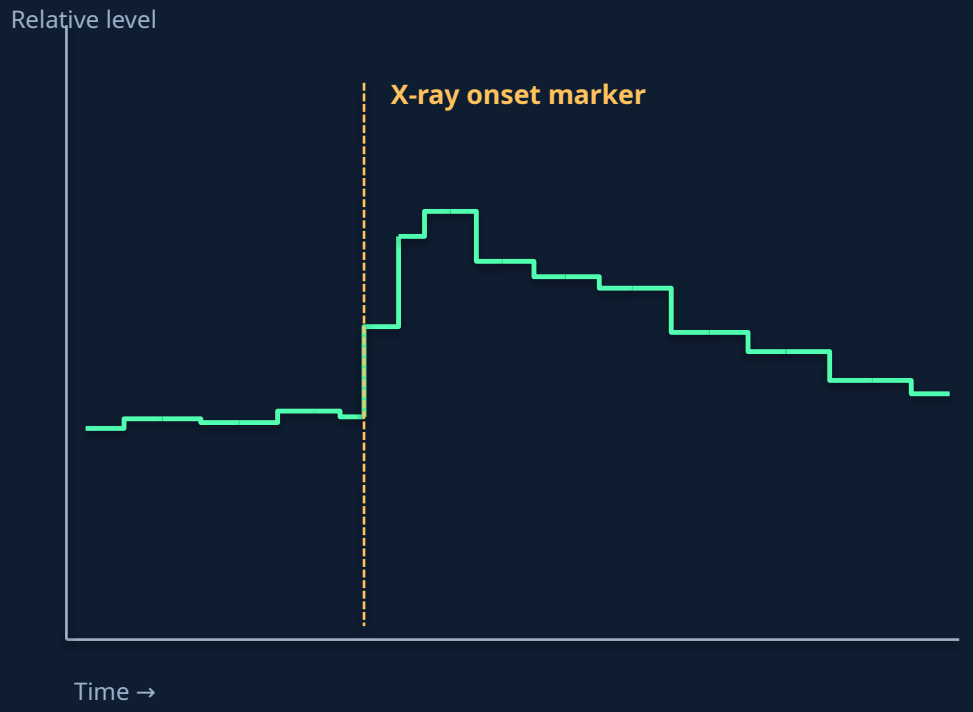
Cosmic Disruption | Sudden Ionospheric Disturbances

SIGNAL INTERPRETATION

CAUSAL CHAIN

01	**Solar flare**	Soft X-rays reach the sunlit atmosphere.
02	**D-region response**	Ionization and electron-neutral collisions change.
03	**Propagation shift**	VLF amplitude and phase may change abruptly.
04	**Cross-check**	Compare timing with GOES X-ray flux; reject local interference.

EXPECTED TRACE — SCHEMATIC ONLY



Direction and magnitude are path-dependent; no fixed amplification is guaranteed.

Analog Frontend | Circuit Architecture

DESIGN REVIEW — VERIFY VALUES AGAINST BENCH MEASUREMENTS

SENSOR
0.62 m telescopic rod + 1.5 m wire

PROTECTION
Series impedance + antiparallel clamps

COUPLING
47 µF series cap blocks DC

GAIN / FILTER
NE5534P stage bias at VREF

PHONE ADC
TRRS mic input if codec supports it

SINGLE-SUPPLY BIAS NETWORK

9 V battery → two matched 10 kΩ resistors → nominal VREF ≈ 4.5 V.

VREF is an AC reference, not a true ground. Bypass it with a suitable capacitor and route the amplifier input/output bias consistently.

NE5534P is not rail-to-rail: available signal swing is less than ±4.5 V.

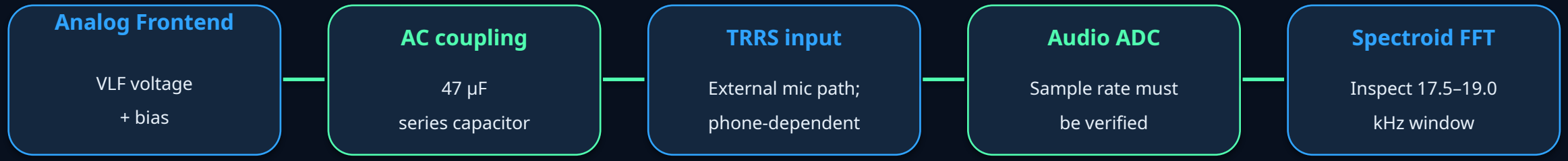
ENGINEERING LIMITS / SAFETY

- **Antenna:** 0.62 m is electrically tiny at λ ≈ 16.5 km; treat as an electric-field pickup, not a resonant monopole.
- **Counterpoise:** Wire changes capacitive coupling and common-mode pickup; it is not a guaranteed inductive ground-plane mirror.
- **Input clamps:** 1N4148 antiparallel clamps need current limiting; they do not eliminate all static/surge risk.

ARCHITECTURE ONLY — confirm op-amp stability, input impedance, bias headroom, protection currents, and phone jack pinout before soldering.

Software & DSP | Smartphone Audio Pipeline

NO LINUX SBC REQUIRED FOR THE FIRST PROTOTYPE



DIGITAL PROCESSING

- **Nyquist:** 44.1 ksample/s gives 22.05 kHz Nyquist; 18.20 kHz is theoretically inside band.
- **Reality:** Codec filters, mic bias, resampling and noise suppression can attenuate or distort 18.2 kHz.
- **Display:** Spectroid is a spectrum viewer, not a calibrated RF instrument.

VALIDATION GATES

- **Verify:** Confirm phone supports external mic input and measure actual audio bandwidth.
- **Measure:** Use a known test tone / reference receiver to confirm the 18.2 kHz response.
- **Avoid:** Do not assume every phone samples at 44.1 kHz or that the room mic is fully bypassed.

Signal chain must be experimentally verified end-to-end; a visible spectral line alone does not prove a calibrated measurement.

System Calibration | Evidence & Cross-Referencing

SEPARATE INSTRUMENT RESPONSE FROM SPACE-WEATHER INTERPRETATION

REPEATABLE VALIDATION PROTOCOL

A	**Carrier baseline**	Record the 18.20 kHz neighborhood over minutes/hours; retain timestamped spectra.
B	**Solar reference**	Compare amplitude/time series with NOAA GOES 1–8 Å X-ray flux.
C	**Station status**	Check VLF monitoring databases (e.g., ILDN/MWLIST where available) for carrier stability/outages.
D	**Counterpoise test**	Connect/disconnect the 1.5 m wire and quantify the line-level change under unchanged gain.
E	**Control condition**	Repeat at different times and with input terminated/shielded to identify local interference.

WHAT COUNTS AS EVIDENCE?

- **Strong:** Repeatable, timestamped amplitude changes that align with GOES flare timing and survive controls.
- **Weak:** A single screenshot or a line that moves when a wire is touched.
- **Caution:** VTX3 carrier frequency may be unstable; track a narrow band, not one exact FFT bin.
- **Limit:** Correlation supports a candidate SID; local interference and receiver gain changes must be excluded.

No flare detection is claimed until field data, controls, and timing correlation are collected.

Conclusion | Optimization & Scalability

PROTOTYPE BOUNDARY AND NEXT ENGINEERING STEPS

RESOURCE-OPTIMIZED DESIGN

- **Compute reuse:** Phone performs sampling, FFT visualization and timestamped observation workflow.
- **Cost target:** Sub-₹500 is a parts-budget target, not a verified final cost or performance claim.
- **Portability:** 9 V battery powers analog frontend; phone supplies display and digital processing.
- **Scientific output:** Initial deliverable is a repeatable VLF carrier monitor; SID classification requires validated data.

SCALABILITY ROADMAP



- **USB audio:** ESP32-S3 may present a USB Audio Class device over native USB, subject to firmware and host compatibility.
- **Logging:** Add timestamped samples, gain metadata, calibration records and CSV export.
- **Upgrade gate:** A dedicated ADC/codec with documented bandwidth is preferable for reliable calibrated VLF measurements.

Engineering claim: low-cost SDR-like analysis is feasible in principle; measured sensitivity, bandwidth and SID performance remain to be demonstrated.